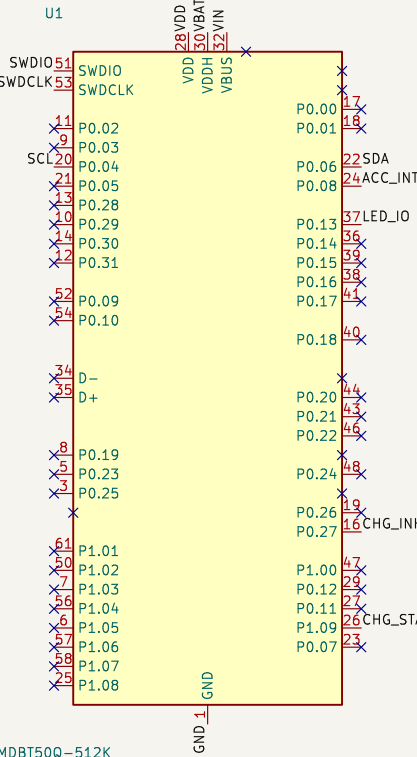


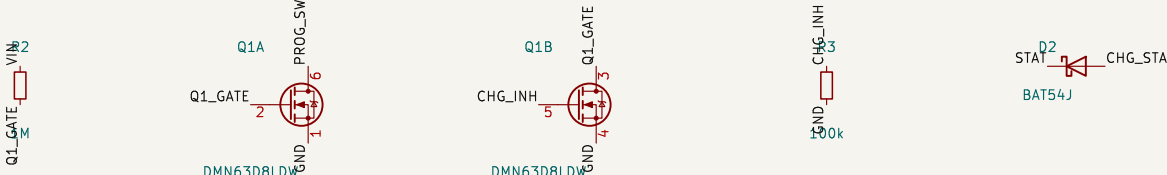
MCU / radio (nRF52833 high-voltage mode)



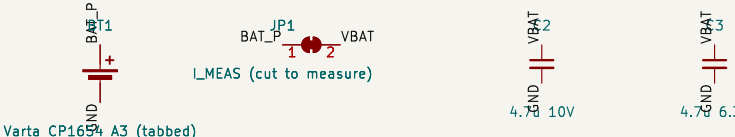
Charging input & charger



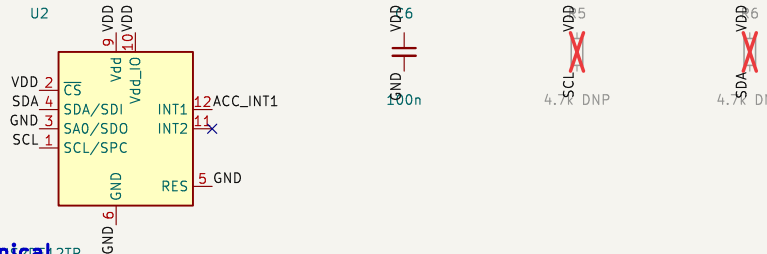
Charge inhibit / status



Battery



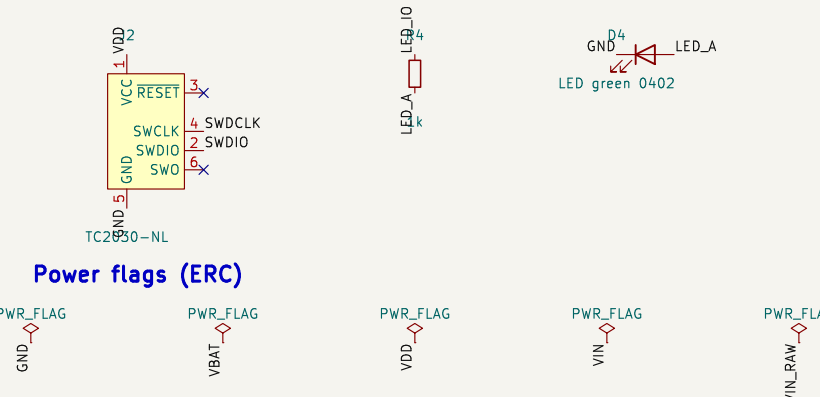
Accelerometer



Mechanical



Programming & status LED



Power flags (ERC)

design.py – ShotPuck circuit, single source of truth.

gen_sch.py turns this into shotpuck.kicad_sch and gen_pcb.py into shotpuck.kicad_pcb. Change the circuit here, then re-run both.

Power path
J1 magnetic pogo (VIN_RAW) -> D1 BAT54J (reverse polarity) -> VIN
VIN -> U3 MCP73831-2 (4.20 V, 50 mA via R1 = 20k) -> VBAT (CP1654)
VBAT -> U1 VDDH (nRF52833 low-voltage mode, REGO -> VDD 3.0 V)
VDD -> U2 LIS2DE12, LED, SWD VCC sense

Charge inhibit (temperature, firmware)
 PROG -> R1 -> Q1A drain; Q1A gate pulled to VIN by R2 (charging allowed whenever a cable is present). Q1B (gate = CHG_INH, R3 pull-down) pulls Q1A gate low -> PROG floats -> MCP73831 charge disabled.

JP1 (bridged solder jumper) in series with the cell: cut, insert an ammeter, re-bridge with solder.

Charge status
STAT -> D2 (cathode) ; D2 anode -> CHG_STAT GPIO with pull-up enabled only
while VBUS is present (no standby leakage).

Generated by hardware/gen_sch.py from design.py

Sheet: /
File: shotpuck.kicad_sch

Title: ShotPuck – arrow shot counter

Size: A3

Size: A3	Date:
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

Rev: A

Id: 1/1